



UNIVERSITY of GUYANA

Ethics, Patents & Intellectual Property in the Caribbean

Sophia Sumner-USI: 1038472

Faculty of Natural Sciences

CSE4201-Ethical Issues and Professional Practice in Computing

Mr. Seon Renville

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Title of Patent: Method & System for managing learning materials presented Offline

Inventor/Applicant: Rouven Krebs & Christian Hochwarth / SAP SE

Patent Number: US 8,644,755 B2

Country of Jurisdiction: United States Patent & Trademark Office

Description: This system covers a system that manages offline learning material for learners.

The invention binds identification strings to parts of the learning content and allows complete tracking of the progress of a learner without the need for connectivity, and subsequently synchronizes the tracking back to a central learning portal once connectivity is regained.

Enrollment, access to content at self-paced rates, and progress reconciliation can be done without live network connectivity in the system.

Introduction

This patent was chosen by me intentionally. My final-year research project is about utilizing a technology intervention to bridge the language gap in linguistically diverse , Guyanese classrooms. In recent years, Guyana has been experiencing an influx of migrants, particularly Venezuelans. The technology intervention features a mobile-first, offline approach to be implemented by teachers that caters to all primary and secondary school classrooms that have linguistically diverse learners, particularly those schools in the hinterland and riverine areas , that are resource strained. My solution is designed to address the exact infrastructural issue that is described in the above patent: How to provide useful, adaptive pedagogical support when there is no reliable broadband, or it is too expensive.

The idea for patent systems is based on an exchange; in return for granting the inventor a monopoly for a limited time, he or she is expected to reveal his or her process to the public (World Intellectual Property Organization [WIPO], 2023). This reasoning also has to some extent been valid even in small developing countries. All of these are outcomes that a student developer would want to achieve if they were looking for national or ministry reach and approach to rollout, and a patented system could achieve all of these. The current structure of the patent system poses structural disadvantages for Caribbean innovators that can't be overlooked. A standard patent-filed application with the USPTO costs a small entity over USD \$1600, and a full prosecution to grant typically requires legal fees somewhere in the tens of thousands of dollars (WIPO, 2023). These figures are really daunting for an independent researcher at a Caribbean university. Moreover, the Patent Cooperation Treaty (PCT) that allows inventors from Guyana to seek international protection from a patent would not require the exact same initial investment. The outcome is a system where big technology companies can file general software patents (patents for methods that Caribbean developers invent independently) and then either impose licensing fees , or threaten litigation against those who don't have the resources to challenge the patent. At this moment it is a structural risk that I have to face personally before publishing my thesis or making my system public in front of the Ministry of Education.

Ethical Concerns

The pattern chosen highlighted three ethical quandaries that were directly relevant to me. Firstly, access and affordability: SAP SE has an active claim for the synchronization of learning

progress offline and for the synchronization of learning progress tracking offline, the exact architecture of my system. While this is indeed intended for humanitarian purposes, a cease and desist could be easy for a guy needing a developer to create a non-commercial version for public schools because there is no educational equity intervention except in IP law for developing context.

Software ownership

In most Caribbean jurisdictions, software is protected by copyright, but there are broad system-level patents in the United States, such as this patent, which are defined as method or apparatus claims. My system has two components; one is the software architecture, and the other is the pedagogical methodology, which is the bilingual schema activation workflow and teacher handbook. Such components would have disparate protection across jurisdictions, giving real legal uncertainty for a developer that is operating regionally.

Fairness towards the small innovator

My research is based on classroom observation in Guyana, collaboration with teachers, and their lived experience. If I publish my thesis before being able to secure IP protection, another actor, with more resources, can come out with a better method and file first. Instead, the pattern system encourages capitalization, and not creativity, or community and betterness, and that is a systemic, not individual, feeling.

Intellectual Property systems in Caribbean countries

IP systems in the Caribbean are not sufficiently effective, but it can be seen. There have been significant achievements in regional harmonization within the framework of CARIFORUM, in capacity building under WIPO Corporation programs (WIPO, 2023). Despite this, the enforcement of these rights is underdeveloped, there is little capacity for a specialist IP judiciary, and the IP awareness of Caribbean professionals in the technology sector is very low. Diana's new domestic rules block sufficient complexity to accommodate software-adjacent inventions, and the only pathway available to student researchers for free commercialization protection is not subsidized. Apology is good, but the infrastructure of the practitioner level hasn't kept up.

Recommendations

The gap can be breached by four measures. First, CARIFORUM needs to have a subsidized pathway to file for student researchers and social innovators. Second, there should be a consistent and planned introduction of IP literacy within the curriculum of computing education at the University of Guyana and other institutions in the region, together with three commercialization audit services. Third, Caribbean governments should promote through the WIPO development agenda on exemption from infringement for non-commercial, publicly funded educational technology. Fourth, open licensing of publicly developed innovation should be required for funding mandates to prevent privatization of community tools. For my research , I am doing the following things: I file a PCT provisional before I submit my thesis. I release the synchronization engine under a Creative Commons non-commercial license and as a family document. Community knowledge is embedded in the handbook, protecting the system and protecting the student for whom it was created.

References

Krebs, R., & Hochwarth, C. (2014). Method and system for managing learning materials presented offline (US Patent No. 8,644,755B2). United States Patent and Trademark Office. <https://patents.google.com/patent/US8644755B2>

World Intellectual Property Organization. (2023). World Intellectual property indicators 2023. WIPO. <https://www.wipo.int/publications/en/details.jsp?id=4678>